LINUX / SYSTEMS

# LVM & RAID Visual Guide

PV/VG/LV layering, snapshots, and RAID 0/1/5/6/10 trade-offs.

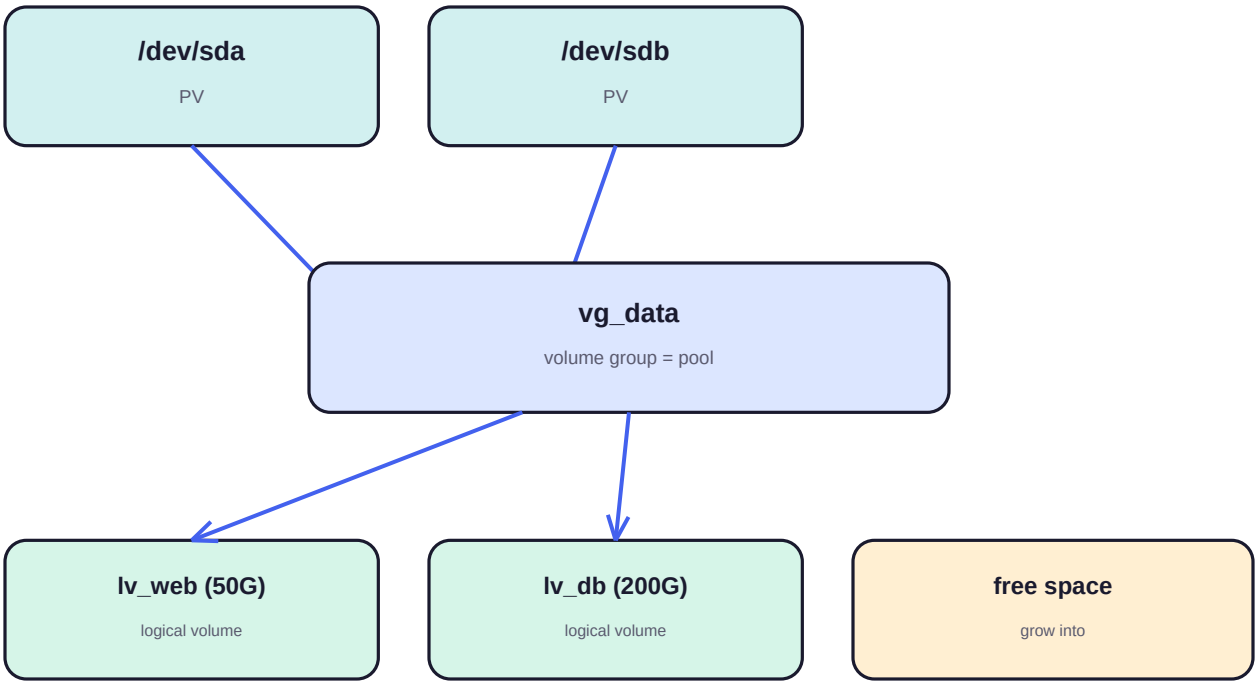
**ANIMATED WALKTHROUGH**

Step through pages to play the animation

STEP 1/6

# LVM Layering: PV → VG → LV

Disks become pools; pools become flexible volumes.

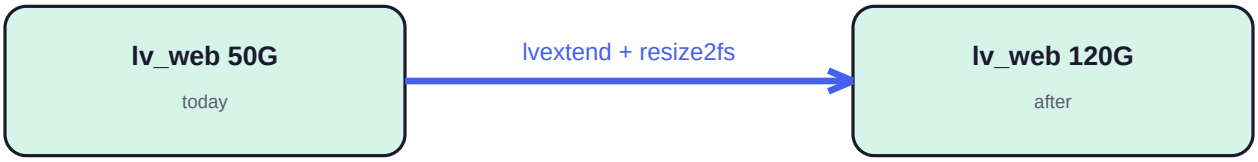



```
$ sudo pvcreate /dev/sdb && sudo vgcreate vg_data /dev/sdb
$ sudo lvcreate -L 50G -n lv_web vg_data
$ sudo mkfs.ext4 /dev/vg_data/lv_web
```

STEP 2/6

# Growing Volumes Online

The superpower: resize storage without downtime.



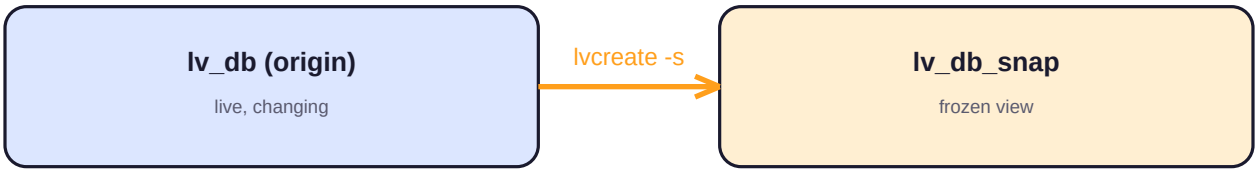

```
$ sudo lvextend -L +70G /dev/vg_data/lv_web   # grow the LV
$ sudo resize2fs /dev/vg_data/lv_web         # grow ext4 (online!)
# xfs: sudo xfs_growfs /mountpoint           # (xfs only grows)
$ sudo vgextend vg_data /dev/sdc             # pool out of space? add disk
```

Order matters: extend the LV first, then the filesystem. Shrinking is the dangerous direction — backup, unmount, e2fsck, resize2fs, lvreduce. ext4 shrinks; XFS never does.

STEP 3/6

# Snapshots: Instant, Copy-on-Write

Freeze a moment for backups — cheap until it changes.




```
$ sudo lvcreate -s -L 10G -n snap /dev/vg_data/lv_db
$ sudo mount -o ro /dev/vg_data/snap /mnt/backup  # backup from here
$ sudo lvremove /dev/vg_data/snap                 # delete when done
```

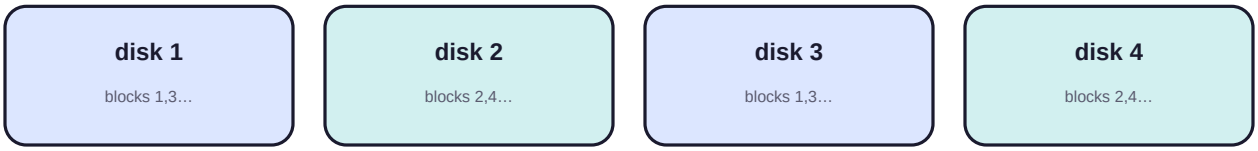
- Snapshots start nearly free; they grow as the origin diverges — size them for the change rate × backup window.
- A full snapshot INVALIDATES itself — monitor snapshot usage during long backups.
- Snapshot + rsync/tar = consistent backup of a live database volume.

STEP 4/6

# RAID 0 & RAID 1: Speed vs Mirror

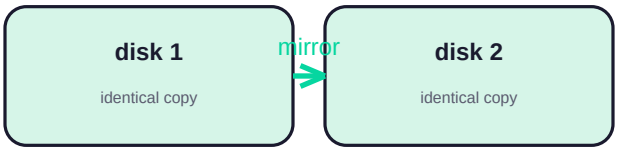
Striping doubles speed and risk; mirroring doubles safety and cost.

### RAID 0 — striping (no redundancy)




1 disk dies → ALL data lost. Capacity: sum of disks.

### RAID 1 — mirroring




1 disk dies → zero data loss. Capacity: ONE disk. Reads can be parallel.

RAID is not backup: it survives DISK failure, not deletion, corruption, or ransomware. You still need snapshots and off-site copies.

STEP 5/6

# RAID 5 / 6 / 10: The Trade-off Table

Parity buys capacity; mirrors buy simplicity.

| Level | Min disks | Survives | Capacity | Use |
|---|---|---|---|---|
| RAID 5 | 3 | 1 disk | (n-1)/n | bulk storage |
| RAID 6 | 4 | 2 disks | (n-2)/n | large arrays |
| RAID 10 | 4 | 1+* | n/2 | databases |

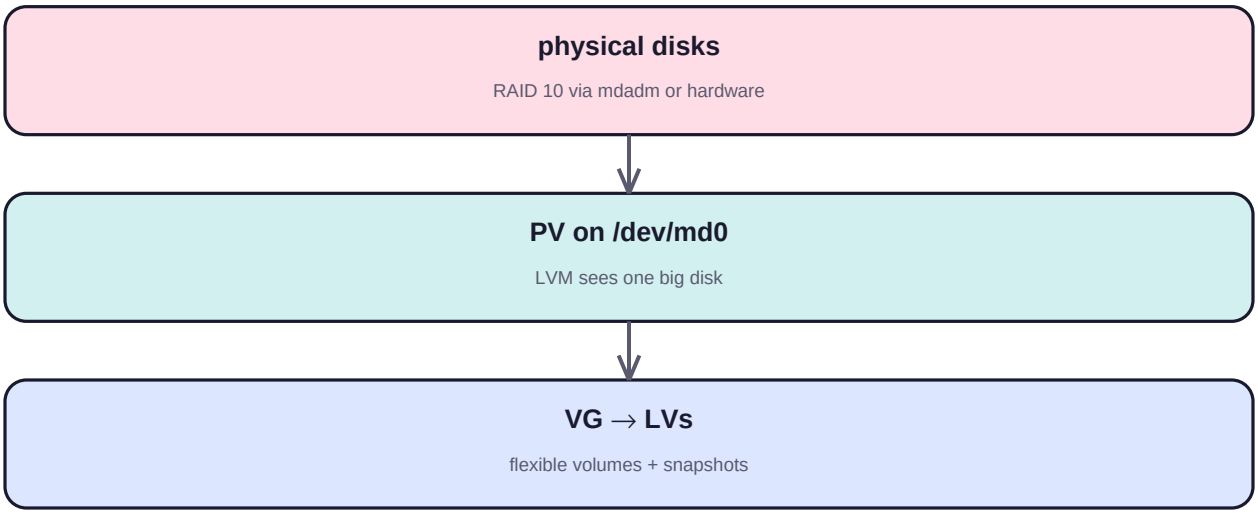
* RAID 10 survives many 2-disk failures — unless both disks of one mirror pair die.

- RAID 5 write penalty: each write = read old data + read parity + write both (the 'write hole' without battery/cache).
- Rebuild times on multi-TB disks stretch into DAYS — during which a second failure kills RAID 5. That's why RAID 6 exists.
- Databases want RAID 10: no parity math on the write path, fastest rebuilds (plain copy).

STEP 6/6

# LVM on RAID: The Standard Stack

Hardware/RAID below, LVM above — each layer does one job.




```
$ sudo mdadm --create /dev/md0 --level=10 --raid-devices=4 /dev/sd[b-e]
$ sudo pvcreate /dev/md0 && sudo vgcreate vg0 /dev/md0
$ cat /proc/mdstat        # watch the sync progress
```

mdadm --detail --scan >> /etc/mdadm/mdadm.conf + update-initramfs -u — or the array won't assemble at boot and nothing above it exists.

STEP ✓/✓

# Key Takeaways

## LVM & RAID

- LVM: PVs (disks) → VG (pool) → LVs (volumes) — grow filesystems online.
- Snapshots are copy-on-write frozen views: perfect for consistent live backups.
- RAID 0 = speed, no safety. RAID 1 = mirror. RAID 5/6 = parity capacity. RAID 10 = DB favorite.
- RAID ≠ backup: it survives disk death, not deletion or ransomware.
- Standard stack: RAID below, LVM above — persist mdadm config or it won't assemble at boot.

**Keep exploring**

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev